

Detailed algorithm for the derivation of the exact solution to the Burgers' equation with our IC & BCs, according to <https://arxiv.org/abs/2409.19240>

Considering the viscous Burgers' equation: $u_t + uu_x = \mathcal{V}u_{xx}$

(1)

Calculate the inverse of the following matrix and name it M^{-1} :

$$M = \begin{pmatrix} u_L & 2\mathcal{V} & 0 & 0 \\ 0 & 0 & u_R & 2\mathcal{V} \\ \frac{1}{2} & \frac{\sqrt{\mathcal{V}}}{2\sqrt{p}} & -\frac{1}{2}e^{-\frac{\sqrt{p}}{\sqrt{\mathcal{V}}}} & -\frac{\sqrt{\mathcal{V}}}{2\sqrt{p}}e^{-\frac{\sqrt{p}}{\sqrt{\mathcal{V}}}} \\ -\frac{1}{2}e^{-\frac{\sqrt{p}}{\sqrt{\mathcal{V}}}} & \frac{\sqrt{\mathcal{V}}}{2\sqrt{p}}e^{-\frac{\sqrt{p}}{\sqrt{\mathcal{V}}}} & \frac{1}{2} & -\frac{\sqrt{\mathcal{V}}}{2\sqrt{p}} \end{pmatrix}$$

(2)

Calculate the following definite integral and name it $A(x, p)$:

$$A(x, p) = \int_0^{0.5} e^{-\frac{y}{2\mathcal{V}}} e^{-\frac{|y-x|\sqrt{p}}{\sqrt{\mathcal{V}}}} dy + \int_{0.5}^1 e^{-\frac{|y-x|\sqrt{p}}{\sqrt{\mathcal{V}}}} dy$$

(3)

Calculate both $A(0, p)$ and $A(1, p)$, meaning that for example once replace x in $A(x, p)$ with 0 and one other time with 1 and calculate the resultant integrals.

(4)

Calculate the derivative of $A(x,p)$ w.r.t. x once, namely $A_x(x,p)$

(5)

take $\omega(0,0)$ as a constant (not important what it is...included for consistency with the paper...will get cancelled later...you can show it with another sign!) and consider the multiplications:

$$R(x,p) = \frac{\omega(0,0)}{2\sqrt{pV}} A(x,p)$$

$$R_x(x,p) = \frac{\omega(0,0)}{2\sqrt{pV}} A_x(x,p)$$

$$R(0,p) = \frac{\omega(0,0)}{2\sqrt{pV}} A(0,p)$$

$$R(1,p) = \frac{\omega(0,0)}{2\sqrt{pV}} A(1,p)$$

(6)

Form the matrix/vector algebra table below and only calculate the value for $U(0,p)$ and $U(1,p)$:

$$(M^{-1}) \cdot \begin{pmatrix} 0 \\ 0 \\ R(0,p) \\ R(1,p) \end{pmatrix} = \begin{pmatrix} U(0,p) \\ U_x(0,p) \\ U(1,p) \\ U_x(1,p) \end{pmatrix} \quad (1)$$

(7)

When the values of $U(0,p)$ and $U(1,p)$ have been calculated, those of $U_x(0,p)$ and $U_x(1,p)$ follow immediately by:

$$U_x(0,p) = -\frac{u_L}{2V} U(0,p)$$

$$U_x(1,p) = -\frac{u_R}{2V} U(1,p)$$

(8)

Now, time to calculate $U(x,p)$:

$$U(x,p) = \frac{1}{2\sqrt{p}} e^{(x-1)\frac{\sqrt{p}}{\sqrt{v}}} [\sqrt{p}U(1,p) + \sqrt{v}U_x(1,p)] + \frac{1}{2\sqrt{p}} e^{-x\frac{\sqrt{p}}{\sqrt{v}}} [\sqrt{p}U(0,p) - \sqrt{v}U_x(0,p)] + R(x,p)$$

(8)

Now, calculate $U_x(x,p)$:

$$U_x(x,p) = \frac{1}{2\sqrt{v}} e^{(x-1)\frac{\sqrt{p}}{\sqrt{v}}} [\sqrt{p}U(1,p) + \sqrt{v}U_x(1,p)] - \frac{1}{2\sqrt{v}} e^{-x\frac{\sqrt{p}}{\sqrt{v}}} [\sqrt{p}U(0,p) - \sqrt{v}U_x(0,p)] + R_x(x,p)$$

(9)

Calculate the inverse Laplace transform of $U_x(x,p)$ or $\mathcal{L}^{-1}\{U_x(x,p)\}$. It should give a function $C(x,t)$!

(10)

Calculate the inverse Laplace transform of $U(x,p)$ or $\mathcal{L}^{-1}\{U(x,p)\}$. It should give a function $D(x,t)$!

(11)

Finally, your exact solution can be retrieved via the formula:

$$u(x,t) = -2\mathcal{V} \frac{C(x,t)}{D(x,t)}$$